



Arrays Problem Bank (Java)

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Strings.

Section 1: Array Basics and Creation

★ **Declare and Initialize a 1D Array**

★ **Access Elements using Index**

★ **Find Length of an Array**

Default Values in Array (int, boolean, String)

Array Declaration Styles (literal vs new)

★ **Multi-Dimensional Array Creation (2D Array)**

★ **Jagged Array Creation (Unequal Row Lengths)**

Array of Objects (Student, Employee)

Section 2: Array Traversal and Display

★ **Print Array Elements using for Loop**

★ **Print Array Elements using Enhanced For-Loop**

★ **Print Array in Reverse**

★ **Print 2D Array (Matrix) Row by Row**

★ **Sum of All Elements in an Array**

★ **Average of Array Elements**

Print Array using Arrays.toString()

Section 3: Searching in Arrays

★ **Linear Search**

★ **Binary Search on Sorted Array**

★ **Find Index of an Element**

Find First and Last Occurrence of an Element

Search in a 2D Array

Find Element Closest to a Given Value

Section 4: Array Sorting

★ **Sort Array in Ascending Order using Arrays.sort()**

★ **Sort Array in Descending Order**



- ★ **Bubble Sort Implementation**
- ★ **Selection Sort Implementation**
 - Insertion Sort Implementation
 - Sort Array of Strings Alphabetically
- ★ **Check if Array is Sorted**

Section 5: Array Aggregation and Statistics

- ★ **Find Maximum and Minimum in an Array**
- ★ **Find Second Largest Element**
 - Find Second Smallest Element
 - Count Even and Odd Numbers in an Array
 - Count Positive, Negative, and Zero in an Array
 - Frequency of Each Element in an Array

Section 6: Array Manipulation

- ★ **Reverse an Array In-Place**
- ★ **Rotate Array Left by K Positions**
 - Rotate Array Right by K Positions
- ★ **Merge Two Arrays**
- ★ **Copy an Array using Arrays.copyOf()**
- ★ **Remove Duplicates from a Sorted Array**
 - Insert Element at a Specific Position
 - Delete Element at a Specific Position

Section 7: 2D Array / Matrix Problems

- ★ **Matrix Addition**
 - Matrix Subtraction
- ★ **Matrix Multiplication**
- ★ **Transpose of a Matrix**
- ★ **Sum of Diagonal Elements**
 - Row-wise and Column-wise Sum
 - Identity Matrix Check
 - Spiral Matrix Traversal

Section 8: Array-Based Logical Problems

- ★ **Check if Array is a Palindrome**
- ★ **Find Missing Number in a Sequence**



★ **Find Duplicate Elements in an Array**

★ **Move All Zeros to End of Array**

★ **Find Pairs with a Given Sum**

Find Intersection of Two Arrays

Find Union of Two Arrays

★ **Maximum Subarray Sum (Kadane's Algorithm)**

Section 9: Real World Array Applications

★ **Student Marks Array Analyzer**

★ **Employee Salary Array Processor**

Inventory Stock Array Manager

★ **Sales Data Array Aggregator**

Attendance Array Tracker